



Figure 3: Environments

that has been built using LXCOS and LXCOS Runner. Details on using Runner can be found in its GitHub pages.

Let's walk through a sample in-house application that has built a provisioning and build release engineering system for Web CMS solutions.

Step 1: Create a project using a template as shown in Figure 1.

With this, we have created a project named *osfy* and the template Drupal7. This means we get a container created with the software packages required by Drupal7, alongside a GitHub repository provisioned with our associated account.

Step 2: Refer to Figure 2.

We have the container details and its IP address in this section.

Step 3: Refer to Figures 3 and 4.

With this, we can have environments (dev, stage, etc) created and access them via an http URL. These are public routable http URLs for accessing the containers. Each environment lets you automatically deploy code from any



Figure 4: Create environments

branch of your GitHub repository.

This is how we build a simple PaaS solution with LXCOS and LXCOS Runner.

There are endless possibilities with LXC in general, and they can be incorporated into LXCOS and LXCOS Runner. We can also contribute more templates (Django, Symphony, etc) and provision them. 

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Changing the disk space for a container:

```
# vzctl stop 101
# vzctl set 101 --diskspace 20G:25G --save
# vzctl stop 101
```

I have specified two values here—20G and 25G. These can be referred to as a soft limit and hard limit, respectively. In order to make the soft limit and hard limit active, you would need to have *vzquota* set to *On*. As *vzquota* itself is a vast topic, it is beyond the scope of this article but you can read more about it at <https://openvz.org/Man/vzquota.8>

Changing the 'root' user password: If you forget the root user password for a container, you can reset it from the host machine. This is true for other users as well but is not needed, because if you have root access to a container, you can reset any user's password.

```
# vzctl stop 101
# vzctl set 101 --userpasswd root:new_password
# vzctl start 101
```

Running any command from the host machine inside the container: This feature allows the administrators of the host machine to run any command inside the container, and the general format is:

```
# vzctl exec CTID command
```

Here are some examples:

- a) `vzctl exec 101 service sshd status`
- b) `vzctl exec 101 df -Th`
- c) `vzctl exec 101 cp -var file1.txt /location/to/copy/`

Gaining access to a container from the host machine:

Sometimes it is necessary to gain access to a container, without the root password, from the host machine. You can do this using the following command:

```
# vzctl enter CTID
```

OpenVZ is a pretty vast topic and if you want to know more, there are unlimited sources on the Internet. The wiki at openvz.org is so comprehensive that you will find all the answers to your doubts there. You can also join the IRC channel (*#openvz*) on the *freenode* network and get assistance from other experienced users, live. 

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